1.5.1 Pressure

Pressure is normal force per unit area. The same force concentrated on a smaller area produces greater pressure. Use the component perpendicular to the surface and the actual area over which it acts. One pascal is one newton of normal force per square metre.

$$p = \frac{F_{\perp}}{A}$$

p: pressure (Pa); F⊥: force normal to the surface (N); A: area (m²).

1.5.2 Boyle's law

For a fixed amount of gas at constant temperature, absolute pressure is inversely proportional to volume. The pressure-volume product is therefore unchanged between two states. Halving volume doubles absolute pressure.

Absolute pressure is measured from a vacuum. If a gauge reads pressure relative to the atmosphere, add local atmospheric pressure to obtain absolute pressure. The two-state comparison requires unchanged gas amount and temperature.

$$p \propto \frac{1}{V}$$
$$pV = \text{constant}$$
$$p_1 V_1 = p_2 V_2$$

Boyle's law: fixed mass of gas at constant temperature. p: absolute pressure; V: volume.

Fixed mass of gas at constant temperature; p: absolute pressure; V: volume. Use consistent units in both states.

1.5.3 Celsius and kelvin temperatures

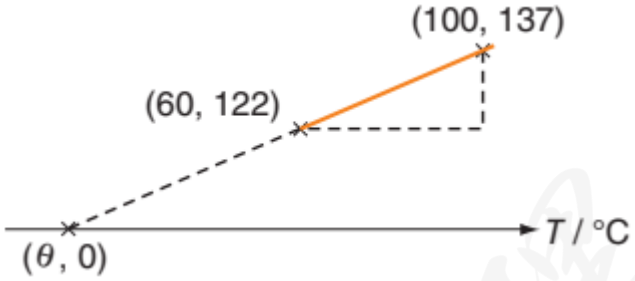
The kelvin scale starts at absolute zero. Add 273.15 to the Celsius reading to obtain the kelvin value. The textbook commonly rounds this to 273. A rise of 1 K equals a rise of 1 °C, but ratios of the readings differ. Gas laws and absolute-temperature proportionalities require kelvin.

$$\frac{T}{\mathrm{K}} = \frac{\theta}{^{\circ}\mathrm{C}} + 273.15 \approx \frac{\theta}{^{\circ}\mathrm{C}} + 273$$

T: thermodynamic temperature; θ: Celsius temperature. The textbook uses the rounded offset 273; 273.15 is exact.

1.5.4 Extrapolating a straight-line graph

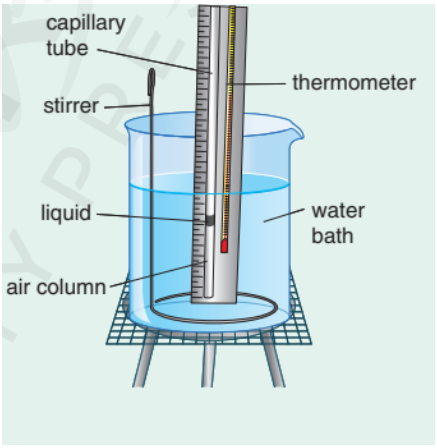
At fixed gas amount and volume, pressure p varies approximately linearly with Celsius temperature θ: $p = m\theta + b$. Two points give the slope $m = \frac{p_2 - p_1}{\theta_2 - \theta_1}$. Extend the line to zero pressure. The intercept $\theta_0 = -\frac{b}{m}$ is the experimental estimate of absolute zero. This is extrapolation beyond the measured range. The textbook data give −265 °C rather than the accepted approximately −273 °C. Calculate from the supplied data and consider experimental error.



Extrapolate the pressure-Celsius-temperature line to zero pressure.

1.5.5 Uniform temperature and steady readings

Stop heating and stir the water bath thoroughly, making its temperature more uniform. Then wait for a steady reading. Stirring reduces temperature differences across the bath. Waiting lets the sample and bath approach thermal equilibrium. Keep the thermometer away from the beaker bottom.



The experimental arrangement includes a water bath, thermometer and gas sample.

1.5.6 Pressure and temperature at constant volume

For a fixed mass of gas at constant volume, absolute pressure is proportional to kelvin temperature. Heating increases the mean momentum transferred to the container walls in collisions. Keep pressure units consistent, convert Celsius readings to kelvin and verify that neither the container volume nor the amount of gas has changed.

$$p \propto T$$

$$\frac{p}{T} = \text{constant}$$

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

Fixed mass and fixed volume; p: absolute pressure; T: temperature in kelvin.

1.5.7 Volume and temperature at constant pressure

For a fixed mass of gas at constant pressure, volume is proportional to kelvin temperature. Heating allows the gas to expand while maintaining its pressure. Compare volume divided by absolute temperature in the two states. Do not use Celsius readings directly or apply the relation when gas leaks or pressure changes appreciably.

$$V \propto T$$

$$\frac{V}{T} = \text{constant}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Fixed mass and constant pressure; V: volume; T: temperature in kelvin.

1.5.8 The combined gas law

For a fixed amount of ideal gas, pV/T remains constant. This relates two states in which pressure, volume and temperature may all change. Here p is absolute pressure measured from a vacuum and T is temperature in kelvin. Use consistent pressure and volume units between states, with no gas entering or escaping.

$$pV \propto T$$

$$\frac{pV}{T} = \text{constant}$$

$$\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$$

Combined gas law for a fixed mass: p is absolute pressure, V is volume, and T is in kelvin. Neither pressure nor volume must be held constant.

Fixed amount of ideal gas; p: absolute pressure; V: volume; T: kelvin temperature.

1.5.9 The ideal gas equation

The ideal gas equation connects absolute pressure p, volume V, amount of gas n and kelvin temperature T. Absolute pressure is measured from a vacuum. With the SI gas constant, use Pa, m^3 and mol. If gas amount changes, apply the equation separately to the initial and final equilibrium states.

The amount in moles is $n = N/N_A = m/M$. N is the number of molecules, N_A is the Avogadro constant, m is total gas mass and M is molar mass. Use matching mass units for m and M, such as g with g mol^{-1} or kg with kg mol^{-1}.

$$pV = nRT$$

p: pressure (Pa); V: volume (m^3); n: amount (mol); T: temperature (K); R = 8.31 J mol^{-1} K^{-1}. Ideal gas.

1.5.10 The microscopic origin of gas pressure

Gas molecules move randomly and change momentum when they collide with container walls, exerting forces on those walls. The mean force from many collisions divided by wall area gives the pressure. Greater molecular number density or greater mean speed can raise pressure. Gas pressure is not caused solely by the gas's weight.

1.5.11 Assumptions of the kinetic theory of gases

The ideal-gas model treats a gas as many identical molecules in continuous random motion. Molecular volume is negligible compared with the space between molecules. Collision duration is negligible compared with the time between collisions. Forces between molecules are ignored except during collisions. Collisions between molecules and with walls are perfectly elastic.

1.5.12 Gas pressure and mean square speed

Kinetic theory relates gas pressure to the number of molecules, the mass of one molecule and mean square speed. It describes the average effect of many collisions with the walls. To find mean square speed, square each molecular speed and then take the average. The model assumes isotropic random motion, meaning the motion has no preferred direction. The molecular form of the equation uses the number of molecules.

$$pV = \frac{1}{3}Nm\overline{c^2}$$

p: pressure (Pa); V: volume (m^3); N: number of molecules; m: mass of ONE molecule (kg). The overbar over c^2 means the mean square speed (m^2 s^{-2}), not the square of the mean speed. Ideal gas with isotropic random motion.

1.5.13 Absolute temperature and mean molecular kinetic energy

The mean translational kinetic energy per ideal-gas molecule depends only on kelvin temperature. Different gases at the same temperature therefore have equal mean translational kinetic energies, although heavier molecules have smaller root mean square speeds. This is an average per molecule, not the total kinetic energy of the whole sample.

$$\langle K\rangle = \frac{1}{2}m\bar{c^2} = \frac{3RT}{2N_{\mathrm{A}}}$$

$$T \propto \langle K\rangle$$

⟨K⟩: mean translational kinetic energy per molecule (J); m: one molecule's mass (kg); mean square speed: $m^2\ s^{-2}$; T: K; R = 8.31 J $mol^{-1}\ K^{-1}$; $N_{\mathrm{A}} = 6.02 \times 10^{23}\ mol^{-1}$.

For an ideal gas, T is absolute temperature (K), and ⟨K⟩ is mean translational kinetic energy per molecule (J).